

CSE331 Quiz 2 Solution
Summer 2025

Let $\Sigma = \{0, 1\}$. Now consider the following languages over Σ .

$L_1 = \{w : 00 \text{ substring appears exactly once in } w\}$

$L_2 = \{w : w \text{ is a palindrome}\}$

$L_3 = \{w : |w| \text{ is even}\}$

$L_4 = L_1 \cap L_2$

a) Write the CFG for the language L_1 . [2 Points]

The language is Regular. The Regular Expression will be:

$(1|01)^* 00 (1|10)^*$

So, we need to just convert this expression to CFG now.

Ans:

$S \rightarrow P 00 Q$

$P \rightarrow 1P \mid 01P \mid \epsilon$

$Q \rightarrow 1Q \mid 10Q \mid \epsilon$

b) Write the CFG for the language L_4 . [3 Points]

In this language, since there is only one occurrence of 00 substring and the strings need to be palindrome, the 00 substring can only be present in the middle and the overall strings can only generate even lengths.

Ans: $S \rightarrow 1S1 \mid 01S10 \mid 00$

c) Write all 7 and 8 letter strings in L_4 . [1 Point]

Ans:

There are no 7 letter strings.

8 Letter Strings: 11100111 , 10100101 , 01100110 .

d) Is $L_3 \cap L_4 = L_4$? Justify your answer. [1 Point]

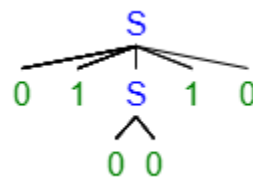
Ans: Yes. Because L_4 automatically generates even length strings.

e) Using the Grammar from b), give a Leftmost Derivation for the String 010010 . [2 Points]

f) Draw a parse tree for the derivation you gave in e). [1 Point]

Ans: e) and f)

S
 $\Rightarrow 01S10$
 $\Rightarrow 010010$



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Let $\Sigma = \{0, 1\}$. Now consider the following languages over Σ .

$L_1 = \{w : 01 \text{ substring appears exactly once in } w\}$

$L_2 = \{w : 10 \text{ substring appears exactly once in } w\}$

$L_3 = \{w : w \text{ is a palindrome}\}$

$L_4 = L_1 \cap L_3$

a) Write the CFG for the language L_1 . [2 Points]

The language is Regular. The Regular Expression will be:

$1^*0^* 01 1^*0^*$

So, we need to just convert this expression to CFG now.

Ans:

$S \rightarrow PQ 01 PQ$

$P \rightarrow 1P \mid \epsilon$

$Q \rightarrow 0Q \mid \epsilon$

b) Write the CFG for the language L_4 . [3 Points]

In this language, since there is only one occurrence of 01 substring and the strings need to be palindrome, then there will be only one occurrence of 10 substring as well.

So, the strings will be like: $0^n 1^m 0^n$ or $1^n 0^m 1^n$; where $n, m \geq 1$.

Ans:

$S \rightarrow P \mid Q$

$P \rightarrow 0P0 \mid 0R0$

$R \rightarrow 1R \mid 1$

$Q \rightarrow 1Q1 \mid 1T1$

$T \rightarrow 0T \mid 0$

c) Write all 7 and 8 letter strings in L_4 that end with 11. [1 Point]

Ans:

7 letter Strings: 1100011, 1110111

8 Letter Strings: 11000011, 11100111

d) Is $L_2 \cap L_3 = L_4$? Justify your answer. [1 Point]

Ans: Yes. Since there is only one occurrence of 01 substring and the strings need to be palindrome, then there will be only one occurrence of 10 substring as well.

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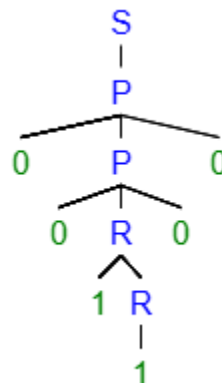
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e) Using the Grammar from b), give a Rightmost Derivation for the String **001100** . [2 Points]

f) Draw a parse tree for the derivation you gave in e). [1 Point]

Ans: e) and f)

S
⇒ P
⇒ 0P0
⇒ 00R00
⇒ 001R00
⇒ 001100



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Let $\Sigma = \{0, 1\}$. Now consider the following languages over Σ .

$L_1 = \{w : 11 \text{ substring appears exactly once in } w\}$

$L_2 = \{w : w \text{ is a palindrome}\}$

$L_3 = \{w : |w| \text{ is odd}\}$

$L_4 = L_1 \cap L_2$

a) Write the CFG for the language L_1 . [2 Points]

The language is Regular. The Regular Expression will be:

$(0|10)^* 11 (0|01)^*$

So, we need to just convert this expression to CFG now.

Ans:

$S \rightarrow P 11 Q$

$P \rightarrow 0P \mid 10P \mid \epsilon$

$Q \rightarrow 0Q \mid 01Q \mid \epsilon$

b) Write the CFG for the language L_4 . [3 Points]

In this language, since there is only one occurrence of 11 substring and the strings need to be palindrome, the 11 substring can only be present in the middle and the overall strings can only generate even lengths.

Ans: $S \rightarrow 0S0 \mid 10S01 \mid 11$

c) Write all 7 and 8 letter strings in L_4 . [1 Point]

Ans:

There are no 7 letter strings.

8 Letter Strings: 00011000 , 01011010 , 10011001 .

d) Is $L_3 \cap L_4 = L_4$? Justify your answer. [1 Point]

Ans: No. Because L_4 only generates even length strings.

e) Using the Grammar from b), give a Rightmost Derivation for the String **101101** . [2 Points]

f) Draw a parse tree for the derivation you gave in e). [1 Point]

Ans: e) and f)

S
 $\Rightarrow 10S01$
 $\Rightarrow 101101$

